1. To write a Python program to add two numbers entered by the user.

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

sum = num1 + num2

print(f"The sum of {num1} and {num2} is {sum}")
```

```
Enter first number: 5
Enter second number: 5
The sum of 5.0 and 5.0 is 10.0
```

2. To write a Python program to interchange the values of two variables.

```
var1 = input("Enter the value for variable 1: ")
var2 = input("Enter the value for variable 2: ")

print(f"\nBefore swapping:\nVariable 1: {var1}\nVariable 2: {var2}")

var1, var2 = var2, var1

print(f"\nAfter swapping:\nVariable 1: {var1}\nVariable 2: {var2}")
```

```
Enter the value for variable 1: 4
Enter the value for variable 2: 2

Before swapping:
Variable 1: 4
Variable 2: 2

After swapping:
Variable 1: 2
Variable 2: 4
```

3. To write a Python program to find the largest among three numbers using conditional statements.

```
num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))
num3 = float(input("Enter the third number: "))

if (num1 >= num2) and (num1 >= num3):
    largest = num1
elif (num2 >= num1) and (num2 >= num3):
    largest = num2
else:
    largest = num3

print(f"The largest number among {num1}, {num2}, and {num3} is: {largest}")
```

```
Enter the first number: 6
Enter the second number: 7
Enter the third number: 8
The largest number among 6.0, 7.0, and 8.0 is: 8.0
```

4. To write a Python program to check whether a given number is even or odd.

```
number = int(input("Enter an integer: "))

if (number % 2) == 0:
    print(f"The number {number} is Even.")
else:
    print(f"The number {number} is Odd.")
```

```
Enter an integer: 7
The number 7 is Odd.
```

5. To write a Python program to determine whether a number is positive, negative, or zero.

```
number = float(input("Enter a number: "))

if number > 0:
```

```
    print(f"The number {number} is Positive.")
elif number < 0:
    print(f"The number {number} is Negative.")
else:
    print(f"The number {number} is Zero.")
```

```
Enter a number: 9
The number 9.0 is Positive.
```

6. To write a Python program to check whether a given year is a leap year.

```
year = int(input("Enter a year: "))


if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print(f"The year {year} is a leap year.")
else:
    print(f"The year {year} is not a leap year.")
```

```
Enter a year: 2026
The year 2026 is not a leap year.
```

7. To write a Python program to perform basic arithmetic operations using operators.

```
num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))

addition = num1 + num2
subtraction = num1 - num2
multiplication = num1 * num2
division = num1 / num2
modulo = num1 % num2
floor_division = num1 // num2


print(f"\nAddition: {num1} + {num2} = {addition}")
print(f"Subtraction: {num1} - {num2} = {subtraction}")
print(f"Multiplication: {num1} * {num2} = {multiplication}")
print(f"Division: {num1} / {num2} = {division}")
print(f"Modulo: {num1} % {num2} = {modulo}")
print(f"Floor Division: {num1} // {num2} = {floor_division}")
```

```
Enter the first number: 19
Enter the second number: 11

Addition: 19.0 + 11.0 = 30.0
Subtraction: 19.0 - 11.0 = 8.0
Multiplication: 19.0 * 11.0 = 209.0
Division: 19.0 / 11.0 = 1.7272727272727273
Modulo: 19.0 % 11.0 = 8.0
Floor Division: 19.0 // 11.0 = 1.0
```

8. To write a Python program to display the multiplication table of a given number.

```
number = int(input("Enter a number to display its multiplication table: "))
print(f"\nMultiplication Table for {number}:")
for i in range(1, 11):
    print(f"{number} x {i} = {number * i}")
```

```
Enter a number to display its multiplication table: 19

Multiplication Table for 19:
19 x 1 = 19
19 x 2 = 38
19 x 3 = 57
19 x 4 = 76
19 x 5 = 95
19 x 6 = 114
19 x 7 = 133
19 x 8 = 152
19 x 9 = 171
19 x 10 = 190
```

9. To write a Python program to calculate the factorial of a number using a loop.

```
number = int(input("Enter a non-negative integer: "))

factorial = 1

if number < 0:
    print("Factorial does not exist for negative numbers.")
elif number == 0:
    print("The factorial of 0 is 1.")
else:
    for i in range(1, number + 1):
        factorial = factorial * i
    print(f"The factorial of {number} is {factorial}.")
```

```
Enter a non-negative integer: 4
The factorial of 4 is 24.
```

10. To write a Python program to find the sum of the first N natural numbers.

```
N = int(input("Enter a positive integer N: "))
sum_natural = 0
if N <= 0:
    print("Please enter a positive integer.")
else:
    for i in range(1, N + 1):
        sum_natural += i
    print(f"The sum of the first {N} natural numbers is: {sum_natural}")
```

```
Enter a positive integer N: 5
The sum of the first 5 natural numbers is: 15
```

11. To write a Python program to print numbers from 1 to N using a for loop.

```
N = int(input("Enter a positive integer N: "))

if N <= 0:
    print("Please enter a positive integer.")
else:
    print(f"Numbers from 1 to {N}:")
    for i in range(1, N + 1):
        print(i)
```

```
Enter a positive integer N: 5
Numbers from 1 to 5:
1
2
3
4
5
```

12. To write a Python program to print numbers in reverse order from N to 1.

```
N = int(input("Enter a positive integer N: "))

if N <= 0:
    print("Please enter a positive integer.")
else:
    print(f"Numbers from {N} to 1 in reverse order:")
    for i in range(N, 0, -1):
        print(i)
```

```
Enter a positive integer N: 7
Numbers from 7 to 1 in reverse order:
7
6
5
4
3
2
1
```

13. To write a Python program to print all even numbers up to a given limit.

```
limit = int(input("Enter a positive integer as the limit: "))
if limit <= 0:
```

```
        print("Please enter a positive integer.")
    else:
        print(f"Even numbers up to {limit}:")
        for i in range(1, limit + 1):
            if i % 2 == 0:
                print(i)
```

```
Enter a positive integer as the limit: 2
Even numbers up to 2:
2
```

14. To write a Python program to reverse the digits of a given number.

```
number = int(input("Enter an integer to reverse: "))

is_negative = False
if number < 0:
    is_negative = True
    number = abs(number)

reversed_num = 0

while number > 0:
    digit = number % 10
    reversed_num = reversed_num * 10 + digit
    number = number // 10

if is_negative:
    reversed_num = -reversed_num

print(f"The reversed number is: {reversed_num}")
```

```
Enter an integer to reverse: 17
The reversed number is: 71
```

15. To write a Python program to count the number of digits in a given integer.

```
number_str = input("Enter an integer: ")

try:
    number = int(number_str)
except ValueError:
    print("Invalid input. Please enter a valid integer.")
else:
    abs_number = abs(number)

    if abs_number == 0:
        num_digits = 1
    else:
        num_digits = len(str(abs_number))

    print(f"The number of digits in {number_str} is: {num_digits}")
```

```
Enter an integer: 96
The number of digits in 96 is: 2
```

16. To write a Python program to calculate the sum of digits of a given number.

```
number = int(input("Enter an integer: "))

abs_number = abs(number)

sum_of_digits = 0

if abs_number == 0:
    sum_of_digits = 0
else:
    temp_num = abs_number
    while temp_num > 0:
        digit = temp_num % 10
        sum_of_digits += digit
        temp_num = temp_num // 10

print(f"The sum of the digits of {number} is: {sum_of_digits}")
```

```
Enter an integer: 567
The sum of the digits of 567 is: 18
```

17. To write a Python program to check whether a given number is prime.

```
num = int(input("Enter an integer to check for primality: "))

if num <= 1:
    print(f"{num} is not a prime number.")
elif num == 2:
    print(f"{num} is a prime number.")
else:
    is_prime = True

    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            is_prime = False
            break

    if is_prime:
        print(f"{num} is a prime number.")
    else:
        print(f"{num} is not a prime number.")
```

```
Enter an integer to check for primality: 12
12 is not a prime number.
```

18. To write a Python program to generate the Fibonacci series up to N terms.

```
num_terms = int(input("Enter the number of terms (N) for the Fibonacci series: "))

a, b = 0, 1

print("Fibonacci Series:")

if num_terms <= 0:
    print("Please enter a positive integer.")
elif num_terms == 1:
    print(a)
elif num_terms > 1:
    print(a)
    print(b)
    for _ in range(2, num_terms):
        next_term = a + b
        print(next_term)
        a = b
        b = next_term
```

```
Enter the number of terms (N) for the Fibonacci series: 8
Fibonacci Series:
0
1
1
2
3
5
8
13
```

19. To write a Python program to check whether a string is a palindrome.

```
text = input("Enter a string: ")
reversed_text = text[::-1]

if text == reversed_text:
    print(f"The string '{text}' is a palindrome.")
else:
    print(f"The string '{text}' is not a palindrome.")
```

```
Enter a string: madam
The string 'madam' is a palindrome.
```

20. To write a Python program to check whether a given number is an Armstrong number.

```
number = int(input("Enter a number: "))
```

```
sum_of_powers = 0
temp_number = number
num_digits = len(str(number))

while temp_number > 0:
    digit = temp_number % 10
    sum_of_powers += digit ** num_digits
    temp_number //= 10

if number == sum_of_powers:
    print(f"The number {number} is an Armstrong number.")
else:
    print(f"The number {number} is not an Armstrong number.")
```

```
Enter a number: 153
The number 153 is an Armstrong number.
```